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T S SPECIFICATION IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE DEPARTMENT OF DEFENSE

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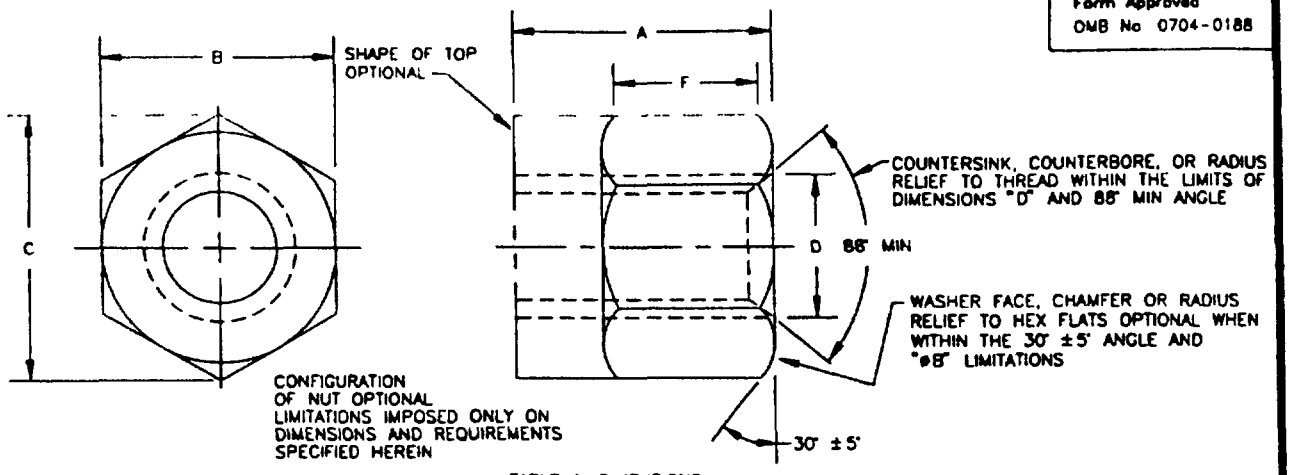


TABLE 1 DIMENSIONS

	DASH NUMBERS NON-METALLIC INSERT AL ALLOY	STEEL	BRASS	CRES	THREAD MIL-S-8879	A MAX	A MIN	(b) B MAX	B MIN	C MIN	D ± 020	F MIN	(a) BEARING SURFACE SQUARENESS
(H)	D02	N02	B02	C02	0860-56UNJC-3B	153	.133	251	243	268	106	044	004
	D04	N04	B04	C04	1120-40UNJC-3B	157	133	251	243	268	136	.044	004
	D06	N06	B06	C06	1380-32UNJC-3B	188	160	313	305	.339	.161	.072	.004
	D08	N08	B08	C08	1640-32UNJC-3B	281	219	345	336	374	188	091	004
	D3	N3	B3	C3	1900-32UNJF-3B	281	208	376	367	410	210	085	004
	D4	N4	B4	C4	2500-28UNJF-3B	.360	240	439	430	482	273	116	005
	D5	N5	B5	C5	3125-24UNJF-3B	.360	271	502	.492	552	336	123	006
	D6	N6	B6	C6	3750-24UNJF-3B	469	318	564	553	622	398	153	006
	D7	N7	B7	C7	4375-20UNJF-3B	.469	.435	690	.679	766	467	178	.007
	D8	N8	B8	C8	.5000-20UNJF-3B	610	575	752	.741	837	531	221	.007
	D9	N9	B9	C9	5625-18UNJF-3B	704	623	877	865	978	594	243	008
	D10	N10	B10	C10	.6250-18UNJF-3B	.766	.732	.940	928	1 051	.656	274	008
	D12	N12	B12	C12	7500-16UNJF-3B	.891	.855	1.064	1.052	1 191	.787	316	.009
	D14	N14	B14	C14	8750-14UNJF-3B	1 016	965	1 252	1 239	1 403	.918	570	.010
	D16	N16	B16	C16	1 0000-12UNJF-3B	1 141	1.010	1 440	1 427	1 615	1 044	635	011
	D18	N18	B18	C18	1 1250-12UNJF-3B	1.266	1.130	1 627	1.614	1.826	1 171	710	012
	D20	N20	B20	C20	1 2500-12UNJF-3B	1 454	1 300	1 815	1 801	2 038	1 295	795	013

(a) BEARING SURFACE SHALL BE SQUARE WITH PITCH DIAMETER WHEN MEASURED IN ACCORDANCE WITH MIL-N-25027
(b) DIMENSION ACROSS FLATS INCLUDE DEFORMATION OF SELF-LOCKING DEVICE

REQUIREMENTS

1 MATERIAL (H) STEEL GRADES 1006 (UNS G10060) PER ASTM A29, ASTM A510 OR ASTM A853
1008 (UNS G10080) PER ASTM A29, ASTM A510 OR ASTM A853
1009 (UNS G10090) PER SAE J118
1010 (UNS G10100) PER AMS 5040, ASTM A29 ASTM A510 OR ASTM A853
1015 (UNS G10150) PER AMS 5060 ASTM A29 ASTM A510 OR ASTM A853
1018 (UNS G10180) PER AMS 5069 ASTM A29 ASTM A510 OR ASTM A853
1035 (UNS G10350) PER AMS 5080, ASTM A29, ASTM A510 OR ASTM A853
1137 (UNS G11370) PER AMS 5024 OR ASTM A29
11L37 (UNS G11374) PER AMS 5020
1213 (UNS G12130) PER SAE J403 OR ASTM A29
12L14 (UNS G12144) PER SAE J403
15821H (UNS H15211) PER AISI 15821H OR SAE J1268

(H) CRES, TYPE 302 (UNS S30200) PER AMS 5688, 303 (UNS S30300) PER AMS 5640, ASTM A581, OR ASTM A582
304 (UNS S30400) PER AMS 5647, 304L (UNS S30430) PER ASTM A493, (UNS S30452) PER ASTM A276

ALUMINUM ALLOY 2014-T6 (UNS A92014) PER QQ-A-200/2
2017-T4 (UNS A92017) PER QQ-A-225/5
2024-T6 (UNS A92024) PER QQ-A-225/6
7075-T6 (UNS A97075) PER QQ-A-200/11

(H) BRASS CA 360 (UNS C36000), 1/2 HARD OR CA 260 (UNS C26000) PER ASTM B36 ASTM B121, ASTM A829, OR AMS 6410

(H) NON-METALLIC INSERT – NYLON PER ASTM D4066, GROUP 1, CLASS 1 OR 2

(H) DENOTES CHANGE(S)

INCH-POUND

PREPARING ACTIVITY DLA-IS
CUSTODIANS ARMY- AR NAVY- AS
AIR FORCE- 99 DLA
REVIEW AS AV CR, MI, SH 8?
USER GL
PROJECT NUMBER 5310-1936
DISTRIBUTION STATEMENT
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MILITARY SPECIFICATION SHEET

TITLE
NUT, SELF-LOCKING, HEXAGON, REGULAR HEIGHT, 250° 125 KSI Fto AND 6?° KSI Fto

SPECIFICATION SHEET NUMBER
MS21044 21 SEP 93 REV H

SUPERSEDING
MS21044 G 20 APR 93

AMSC- N/A FSC 5310

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2 PROTECTIVE COATING. STEEL AND BRASS NUTS - CADMIUM PLATED IN ACCORDANCE WITH QQ-P-416 TYPE II CLASS 2 (H)
CORROSION-RESISTANT STEEL NUTS - PASSIVATE IN ACCORDANCE WITH QQ-P-35
ALUMINUM ALLOY NUTS - ANODIZE OR CHEMICALLY SURFACE TREAT IN ACCORDANCE WITH PROCUREMENT SPECIFICATION
ALUMINUM ALLOY NUTS - IN SIZES THRU 0 2500-28 DYED BLUE PER PROCUREMENT SPECIFICATION

3 LUBRICANT. LUBRICANT APPROVED IN ACCORDANCE WITH MIL-N-25027, LUBRICANT SHALL BE SOLUBLE IN THE CLEANER SPECIFIED IN THE PROCUREMENT SPECIFICATION

4 MAGNETIC PERMEABILITY. MAGNETIC PERMEABILITY OF CORROSION-RESISTANT STEEL NUTS SHALL BE LESS THAN 2 0 (AIR=1 0) FOR A FIELD STRENGTH H = 200 OERSTEDS WHEN TESTED IN ACCORDANCE WITH ASTM A342 METHOD 6 (H)

5 SURFACE ROUGHNESS. SURFACE ROUGHNESS IN ACCORDANCE WITH ANSI/ASME B46 1 UNLESS OTHERWISE SPECIFIED, SURFACES SHALL NOT EXCEED 125 MICROINCHES

6 VIBRATION LIFE. MINIMUM VIBRATION LIFE REQUIREMENT FOR NON-METALLIC INSERT PARTS SHALL BE THREE TIMES AVERAGE VIBRATION LIFE LISTED IN MIL-N-25027

7 PART NUMBER. THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I

EXAMPLE OF PART NUMBERS

MS21044N4 = 2500-28 STEEL NUT NON-METALLIC INSERT, CADMIUM PLATED
MS21044D4 = 2500-28 ALUMINUM ALLOY NUT, NON-METALLIC INSERT, ANODIZED
MS21044B4 = 2500-28 BRASS NUT, NON-METALLIC INSERT, CADMIUM PLATED
MS21044C4 = 2500-28 CRES NUT, NON-METALLIC INSERT PASSIVATED

TABLE II PERFORMANCE CHARACTERISTICS

THREAD MIL-S-8879	AXIAL STRENGTH (LB/MIN)			WEIGHT, LB/100 REF		
	STEEL	AL ALLOY	CRES	AL ALLOY	STEEL	BRASS
(H) 0860-56UNJC-3B	440	220	440	06	15	16
1120-40UNJC-3B	750	750	750	06	15	16
1380-32UNJC-3B	1,130	1,130	1,130	11	30	32
1640-32UNJC-3B	1 720	1,720	1,720	18	50	54
1900-32UNJF-3B	2 460	2,460	2,460	.21	58	.63
2500-28UNJF-3B	4 580	4,580	4,580	36	99	1 10
3125-24UNJF-3B	7 390	3,670	7,390	50	1 40	1 50
3750-24UNJF-3B	11,450	5,680	11,450	78	2 15	2 30
4375-20UNJF-3B	15,450	7,660	15,450	1 20	3 40	3 70
5000-20UNJF-3B	21 110	10,470	21 110	1 70	4 70	5 10 (H)
.5625-18UNJF-3B	26,810	13,330	26,810	2 80	7 80	8 50
6250-18UNJF-3B	34 130	16,930	34,130	3 40	9 50	10 30
7500-16UNJF-3B	50 020	24,810	50 020	5 00	13 60	15 00
8750-14UNJF-3B	68,440	33,950	68 440	7 60	21 00	23 00
1 0000-12UNJF-3B	92,180	45,720	92 180	11 00	31 00	33 00
1 1250-12UNJF-3B	116 700	57,880	116,700	17 00	46 00	50 00
1 2500-12UNJF-3B	147,940	73,380	147,940	24 00	65 00	70 00

NOTES.

1 BREAK ALL SHARP EDGES AND REMOVE ALL BURRS AND SLIVERS

2 DIMENSIONS ARE IN INCHES.

3 DESIGN AND USAGE LIMITATIONS THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTH OF BOLTS AND SCREWS WITH AN ULTIMATE STRESS (Ftu) OF 125 KSI (60 KSI FOR ALUMINUM PARTS) BASED ON THE CROSS-SECTIONAL AREA AT THE BASIC ROOT DIAMETER OF THE THREAD THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNAL THREADS PER MIL-S-8879 THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588

4 INTERCHANGEABILITY RELATIONSHIPS (STATUS NOTE) MS21044 NUTS CAN UNIVERSALLY REPLACE AN365, MS20365 AND NAS1021 NUTS OF LIKE MATERIAL PLATING AND THREAD SIZE, EXCEPT FOR AN365 AND MS20365 7/16-20 NUTS NON-METALLIC INSERT LOCKING DESIGNS CAN BE SUBSTITUTED FOR THE INACTIVATED BRASS OR ALUMINUM ALLOY ALL-METAL LOCKING DESIGN NUTS AN365, MS20365 AND NAS1021 NUTS CANNOT UNIVERSALLY REPLACE THE MS21044 NUTS FOR DEFINITION AND APPLICATION OF DRAWING STATUS NOTES SEE MIL-STD-962

5 IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE

6 REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN

7 BIDS SHALL BE SOLICITED ONLY FROM THE MANUFACTURERS OR DISTRIBUTORS LISTED ON QPL-25027

PREPARING ACTIVITY DLA-IS
CUSTODIANS ARMY- AR NAVY- AS
AIR FORCE 99 DLA
REVIEW AS AV, CF MI, SH, 82
USER GL
PROJECT NUMBER 5310-1936
DISTRIBUTION STATEMENT A Approved for public release distribution is unlimited

MILITARY SPECIFICATION SHEET

TITLE NUT SELF LOCKING, HEXAGON, REGULAR HEIGHT, 250°F, 125 KSI Ftu AND 60 KSI Ftu

SPECIFICATION SHEET NUMBER MS21044 21 SEP 93 REV H
SUPERSEDING MS21044 G 20 APR 93
AMSC N/A FSC 5310

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TABLE III INTERCHANGEABILITY

SEE INTERCHANGEABILITY RELATIONSHIP STATUS NOTE												SUBSTITUTE PART NUMBERS
CANCELLED PART NUMBERS												
AN365			MS20365			NAS1021		MS21044				
-440	-440C	-440A	-440	-440C	-440A	NO4	---	---				MS21044 N04
B440	B440C	B440A	B440	B440C	B440A	B04	E04	E04				B04
D440	D440C	D440A	D440	D440C	D440A	D04	H04	H04				D04
-632	-632C	-632A	-632	-632C	-632A	N06	---	---				N06
B632	B632C	B632A	B632	B632C	B632A	B06	E06	E06				B06
D632	D632C	D632A	D632	D632C	D632A	D06	H06	H06				D06
-832	-832C	-832A	-832	-832C	-832A	N08	---	---				N08
B832	B832C	B832A	B832	B832C	B832A	B08	E08	E08				B08
D832	D832C	D832A	D832	D832C	D832A	D08	H08	H08				D08
-1032	-1032C	-1032A	-1032	-1032C	-1032A	N3	--	--				N3
B1032	B1032C	B1032A	B1032	B1032C	B1032A	B3	E3	E3				B3
D1032	D1032C	D1032A	D1032	D1032C	D1032A	D3	H3	H3				D3
-428	-428C	-428A	-428	-428C	-428A	N4	--	--				N4
B428	B428C	B428A	B428	B428C	B428A	B4	E4	E4				B4
D428	D428C	D428A	D428	D428C	D428A	D4	H4	H4				D4
-524	-524C	-524A	-524	-524C	-524A	N5	--	--				N5
B524	B524C	B524A	B524	B524C	B524A	B5	E5	E5				B5
D524	D524C					D5	H5	H5				D5
-624	-624C	-624A	-624	-624C	-624A	N6	--	--				N6
B624	B624C	B624A	B624	B624C	B624A	B6	E6	E6				B6
D624	D624C	D624A	D624	D624C	D624A	D6	H6	H6				D6
-720	-720C	-720A	-720	-720C	-720A	N7	--	--				N7
B720	B720C	B720A	B720	B720C	B720A	B7	E7	E7				B7
D720	D720C	D720A	D720	D720C	D720A	D7	H7	H7				D7
-820	-820C	-820A	-820	-820C	-820A	N8	--	--				N8
B820	B820C	B820A	B820	B820C	B820A	B8	E8	E8				B8
D820	D820C	D820A	D820	D820C	D820A	D8	H8	H8				D8
(H) -918	-918C	-918A	-918	-918C	-918A	N9	--	--				N9
B918	B918C	B918A	B918	B918C	B918A	B9	E9	E9				B9
D918	D918C	D918A	D918	D918C	D918A	D9	H9	H9				D9
-1018	-1018C	-1018A	-1018	-1018C	-1018A	N10	---	---				N10
B1018	B1018C	B1018A	B1018	B1018C	B1018A	B10	E10	E10				B10
D1018	D1018C	D1018A	D1018	D1018C	D1018A	D10	H10	H10				D10
-1216	-1216C	-1216A	-1216	-1216C	-1216A	N12	---	---				N12
B1216	B1216C	B1216A	B1216	B1216C	B1216A	B12	E12	E12				B12
D1216	D1216C	D1216A	D1216	D1216C	D1216A	D12	H12	H12				D12
-1414	-1414C	-1414A	-1414	-1414C	-1414A	N14	---	---				N14
B1414	B1414C	B1414A	B1414	B1414C	B1414A	B14	E14	E14				B14
D1414	D1414C	D1414A	D1414	D1414C	D1414A	D14	H14	H14				MS21044 D14
-1614	-1614C	-1614A	-1614	-1614C	-1614A	---	---	---				NAS1021 N17
B1614	B1614C	B1614A	B1614	B1614C	B1614A	---	E17	---				NAS1021 B17
D1614	D1614C	D1614A	D1614	D1614C	D1614A	---	H17	---				NAS1021 D17
---	---					N16	---	---				MS21044 N16
---	---					B16	E16	E16				B16
---	---					D16	H16	H16				D16
-1812	-1812C	-1812A	-1812	-1812C	-1812A	N18	---	---				N18
B1812	B1812C	B1812A	B1812	B1812C	B1812A	B18	E18	E18				B18
D1812	D1812C	D1812A	D1812	D1812C	D1812A	D18	H18	H18				D18
-2012	-2012C	-2012A	-2012	-2012C	-2012A	N20	---	---				N20
B2012	B2012C	B2012C	B2012	B2012	B2012C	B20	E20	E20				B20
D2012	D2012C	D2012A	D2012	D2012C	D2012A	D20	H20	H20				MS21044 D20

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CUSTODIANS ARMY-AR NAVY-AS
AIR FORCE-99 DLA-
REVIEW AS, AV, CF, MI, SH, 82
USER CL
PROJECT NUMBER 5310-1936

MILITARY SPECIFICATION SHEET
TITLE
NUT, SELF-LOCKING HEXAGON, REGULAR, HEIGHT, 250"F, 125 KSI Ftu AND 60 KSI Ftu

SPECIFICATION SHEET NUMBER
MS21044 21 SEP 93 REV H
SUPERSEDING
MS21044 G 20 APR 93
AMSC N/A FSC 5310

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